

THOMAS SLAVONIA

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EDUCATION

University of California, San Diego Master of Science, Electrical Engineering	Sep 2025 - June 2027
University of California, Los Angeles Bachelor of Science, Mathematics	Aug 2020 - June 2024
Relevant Coursework: Planning and Learning in Robotics, Linear Systems, Statistical Learning Theory II, Sensing and Estimating Robotics, Real Analysis, Enumerative Combinatorics, Algorithms, Applied Numerical Methods, Probability Theory (Graduate), Graph Theory, Machine Learning, Fourier Analysis, Complex Analysis (Honors), Data Structures and Algorithms	

TECHNICAL SKILLS

Programming/Development Tools: Python, C++, Lua, Bash, Neovim, Git, LaTeX

Libraries: NumPy, PyTorch, Pandas, Scikit-Learn, Matplotlib, SciPy, CasADI

Software & Platforms: Excel (Advanced: VBA, Power Query, Power Pivot, API Integration), Google Suite, Premiere Pro

PROJECTS

Motion Planning with ARA* Designed my own implementation of ARA* to navigate 3-D environments with static and moving targets	May 2026 - May 2026
Waymo LiDAR Semantic Segmentation - Group project with two teammates where we implemented and benchmarked four point-cloud model backbones (PointNet, EdgeConv, PointNet++, handcrafted geometric features) and two range-image backbones (UNet, cross-attention UNet) with a modular design for easy extension - Developed a data processing pipeline to convert the raw Waymo LiDAR data into point clouds and created unique visualizations of the point clouds for classification verification - Tested each model backbone on the Waymo Open LiDAR dataset	Feb 2026 - March 2026
Visual Inertial SLAM - Derived and implemented EKF prediction using discrete-time pose kinematics with IMU measurements. - Built an EKF landmark update step using a stereo camera observation model; computed a stereo baseline from extrinsic calibration matrices and constructed the stereo projection matrix from scratch - Implemented full visual-inertial SLAM combining joint pose and landmark state estimation with a sparse covariance matrix using SciPy sparse matrices	Feb 2026 - March 2026
3-D Camera Orientation Tracking and Panorama Recreation - Implemented a projected gradient descent algorithm to track camera orientation using IMU sensor data, quaternion kinematics, and formulated a constrained optimization problem over unit quaternion trajectories using PyTorch - Reconstructed panoramic images by stitching time-synchronized RGB camera frames using estimated orientation trajectories, mapping pixel coordinates across projection surfaces	Jan 2026 - Feb 2026

EXPERIENCE

INCEPTION Lab, San Diego, CA: Research Intern - Developed Python simulations for control theory research papers in the areas of nonlinear and hybrid dynamical systems - Interpreted novel control theory formulas and research into simulation code in Python, designing animated visualizations using Matplotlib - Current work is creating simulations in the context of multi-agent systems in Nash equilibrium-seeking deceptive games	Apr 2026 - Present
Build Group, San Francisco, CA: Estimator Intern - Developed predictive cost models for construction materials, equipment, and personnel using regression analysis and statistical forecasting within Excel - Built and implemented automated ETL pipelines using Excel VBA, Power Query, and API integrations to connect third-party estimating software with internal databases - Led enterprise-wide adoption of new third-party software solutions by developing custom Excel interfaces, macros, and data transformation workflows for seamless team integration - Developed structural engineering calculations to approximate the needed material for the given project	June 2025 - Sept 2025
Mathematics Directed Reading Program - Selected to participate in a directed reading program mentored by a graduate student studying connections between Young Tableaux and the representation theory of the symmetric group - Studied advanced mathematical concepts related to Pipe Dreams and Schubert Calculus alongside a graduate student who was researching questions related to Pipe Dreams - Presented algebraic combinatorics material to a graduate student audience and effectively addressed challenging questions	Jan 2024 - June 2024
Data Structures and Algorithms Learning Assistant - Led weekly discussion sections and office hours for the Data Structures and Algorithms course at UCLA, assisting students with projects - Presented workshops training students in fundamental concepts and held review sessions prior to exams to enhance student performance	Dec 2023 - June 2024